

Integration by Substitution

Today's Goals

- ◆ You should be able to explain the process of integration by substitution in relation to the Chain Rule.
- ◆ You should be able to use substitution to evaluate definite and indefinite integrals.
- ◆ You should be able to determine whether a particular substitution is helpful in evaluating an integral.

1. Can you evaluate the following integrals?

(a) $\int_1^4 \frac{1}{3} \sqrt{u} \, du$

(b) $\int \cos(3x) \, dx$

(c) $\int \cos(x^2) \, dx$

2. Fill in the blank in each statement:

(a) $\int \underline{\hspace{2cm}} \, du = \sin(u) + C$

(b) $\int \underline{\hspace{2cm}} \, dx = \sin(x^2) + C$

Integration by Substitution

If f is a continuous function with antiderivative F and g is a differentiable function, then

$$\int f(g(x)) \cdot g'(x) \, dx = \underline{\hspace{2cm}}.$$

We often write $u = g(x)$ and $du = g'(x) \, dx$, in which case the formula is

$$\int f(g(x)) \cdot g'(x) \, dx = \underline{\hspace{2cm}}.$$

3. Evaluate the following integrals using substitution.

(a) $\int \frac{\sin(\ln x)}{5x} dx$

(b) $\int e^{-2x} \cos(e^{-2x}) dx$

(c) $\int_1^4 x^2 \sqrt{1+x^3} dx$

(d) $\int \sin x \cos^5 x dx$

(e) $\int_{1/e}^{e^3} \frac{4 \ln x}{x} dx$

(f) $\int \frac{e^x}{2e^x + 5} dx$

(g) $\int_0^1 \sqrt{5x + 4} dx$

4. One of the following integrals can easily be evaluated using substitution. Which one? Evaluate it.

(a) $\int \frac{1}{e^{x^2}} dx$

(b) $\int \frac{x}{e^{x^2}} dx$

5. Suppose g is a mysterious function, and the only thing you know about it is that $\int_0^{12} g(x) dx = \pi$. For each of the following integrals, decide whether you have enough information to evaluate the integral; if so, evaluate it.

(a) $\int_0^{12} g(3x) dx$

(b) $\int_0^4 g(3x) dx$

(c) $\int_0^{144} g(x^2) dx$

(d) $\int_0^{2\sqrt{3}} xg(x^2) dx$

(e) $\int_0^{144} xg(x^2) dx$